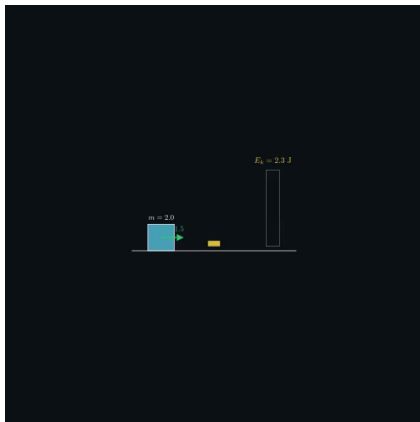


Motivation: Why Multiple Variables?

- In single-variable calculus, we study $y = f(x)$ (one input causes one output).
- In real-world physics, data science, and IT, quantities depend on **several independent inputs**.



Non-linear scaling: $E_k \propto m$ and $E_k \propto v^2$

Physical Examples

- **Kinetic energy:**

$$E_k = \frac{1}{2}mv^2$$

(m is mass, v is velocity).

- **Potential energy:**

$$E_p = mgh$$

(m is mass, h is height).